

VIVEKANAND B

Engineer • Intrapreneur • AI-Native Architect
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SUMMARY

Staff Software Engineer and Technical Lead with an intrapreneurial DNA and a multidisciplinary foundation spanning Mechanical, Electronics, and Software Engineering. Expert in applying the precision of aerospace design (ISRO, Safran) to build scalable, privacy-first, enterprise-grade AI products. Currently driving "Zero-to-One" initiatives at Sanas.ai, operating as an internal technical co-founder to deliver the Sanas Consumer App, Sales Copilot, edge-powered Speech Intelligence platforms, and high-throughput audio orchestration pipelines from concept to GA. A pioneer in AI-Native Development, orchestrating multi-agent workflows to accelerate delivery timelines by 40%+ while bridging the gap between Research, Engineering, and Product strategy.

SKILLS

- **AI & LLM:** RAG, Vector Databases, Prompting, Agentic Workflows (Cursor/Claude/Codex).
- **Mobile & Frontend:** React, React Native (Expo), NativeWind, TypeScript, Next.js, Vue.js
- **Backend & Architecture:** Python (FastAPI), Nest/Node.js, Microservices, WebSocket/WebRTC.
- **Cloud & ML/DevOps:** GCP/AWS, Kubernetes, Docker, Terraform, MLflow/Kubeflow/Ray CI/CD, Clickhouse
- **Core Engineering:** CAD/CAM (CATIA), Finite Element Analysis (Ansys), Industrial Automation.

EXPERIENCE

PROGRAMMING

Staff Engineer | Sanas.ai | Bangalore, India | Nov 2024 – Present

Software Engineer | NovaSignal/NeuraSignal | Nov 2020 – Nov 2024

- **Speech Intelligence (SI) Platform:** Architected & Implemented a privacy-first, edge-powered voice analytics platform to process high-fidelity audio streams locally without cloud PII exposure; engineered a ClickHouse OLAP pipeline for incremental metric aggregation, queue-driven multi-track LLM workers for automated post-call CRM summaries, and a multi-tier presentation layer featuring a real-time active call widget and a 30-second refresh fraud dashboard.
- **Playground:** Designed and implemented an ElevenLabs-style interactive web platform and a triple-track asynchronous audio orchestration pipeline (FastAPI/Asyncio) to concurrently run Accent Translation, Noise Cancellation, and Language Translation within a single session; optimized inference on NVIDIA Triton servers via gRPC to maintain sub-100ms chunk latency, automated auto-scaling across AWS EKS and Modal (H100/T4 GPUs) to handle 430k requests daily, and engineered an eager-ingestion speculative processing layer for background voice cloning.
- **Sanas Consumer Mobile App (React Native):** Built and shipped a production-ready B2C app in <4 weeks with zero prior mobile experience by orchestrating a multi-agent AI workflow (Cursor/Claude). Achieved 100% iOS/Android parity (handling complex permissions) and implemented localization for 9 languages, reducing user onboarding time to ~1.5 minutes.
- **Sanas for Sales (Real-Time AI Copilot):** Architected a hybrid Chrome Extension/Desktop solution from concept to Enterprise GA in 3 months. Led the technical roadmap and weekly C-suite reviews, engineering a multi-modal RAG pipeline with <4s latency for live objection handling.
- **Unified ML Platform (Infrastructure):** Led a cross-functional team (DevOps/MLOps/Innovation) to consolidate fragmented tooling into a centralized Kubernetes-native ML Platform. Conducted technical audits with 10+ Research Scientists to design a "Hub-and-Spoke" architecture using MLflow, resolving reproducibility issues and accelerating model deployment.
- **AI-Next Strategy (Enterprise Adoption):** Spearheaded the organizational shift from "AI Users" to "Agent Directors." Established a vendor-agnostic toolchain and investment dashboard, democratizing access to advanced coding agents for 80% of the workforce and defining the "Crawl, Walk, Run" adoption strategy.

- **Admin Portal (Core Engineering):** Designed and implemented a workspace-level Time Zone Management System & App Version Distribution, enabling consistent global reporting standards across India, Philippines, and US regions, reducing support tickets by 60%.
- **Cloud Architecture & Telemetry (GCP):** Designed a centralized, serverless telemetry architecture using **Cloud Run** and **BigQuery** to replace manual diagnostics, enabling real-time fleet monitoring and predictive maintenance for robotic systems. Built the **Data View** application (Streamlit/Flask) to visualize hospital utilization metrics, empowering sales and leadership with actionable business intelligence.
- **AI-Driven Automation & Tooling:** Pioneered the integration of **OpenAI APIs** into QA workflows (AutoUI, TestAI), automating test script generation and reducing regression cycles by **30%**. Developed an AI-powered **Jira Analyzer** to classify issues and visualize error paths, streamlining system performance analysis.
- **Healthcare Interoperability & Microservices:** Engineered secure cloud microservices (Jupiter, Venus releases) compliant with **HIPAA**, **HL7**, and **DICOM** standards to ensure seamless data exchange between robotic devices and hospital EMR systems. Developed automated compliance tools, including a DICOM Decoder and SOUP Creator, to validate medical imaging data and audit open-source security dependencies.

ENTREPRENEURIAL

Founder & Technician | Gadjoy Repair Services | Bangalore, India
Nov 2016 – Nov 2020

- Founded and managed a successful laptop and gadget repair service, handling over 1,000 devices and 100+ customers per month, and achieving a 4.7+ customer satisfaction rating across multiple platforms.
- Developed and implemented custom software systems to monitor customer devices, manage checklists, and streamline ledger and enquiry entries. Leveraged Excel as a functional database, utilizing advanced features to trace device activities, which increased operational efficiency by 80%.
- Acquired transferable skills in team management, customer service, and business development, alongside technical expertise in software development, database management

MECHANICAL

Senior Lead Engineer | Legend Technologies | Jan 2013 – Nov 2016

Trainee Engineer | Safran Engineering India | Aug 2011 – Dec 2012

- Led multidisciplinary teams in the design and fabrication of aerospace & locomotive jigs and fixtures for clients like ISRO, Safran, and Pratt & Whitney.
- Provided end-to-end support in manufacturing, testing, and service, overseeing the entire product lifecycle
- Utilized advanced CAD and simulation tools to do mechanical design and analysis of aerospace components and developed custom software scripts to automate design tasks reducing design time & in the process gaining transferable skills like applying analytical and project management.

EDUCATION

- **Bachelor of Engineering (Mechanical Engineering)** - Visvesvaraya Technological University (VTU) | First Class with Distinction | 2011
- Deep Learning Specialization & Machine Learning Specialization (Andrew Ng) & Multiple Short Course on AI, Tooling, Ops

PATENTS & PUBLICATIONS

- **Patents:** System and Method of Generating Image of Vascular Flow Network - US20230329668A1
- **White Papers:** Published research on aerospace manufacturing and slip ring design.

VOLUNTEERING

- Contributed to the development of an IoT-based air quality monitoring system ([AirCare](#)) and a COVID relief distribution platform ([Stop Hunger](#)) with mapshalli.org, Whitefield, Bangalore.
- Mentored school children, engineering graduates (for skill development), and junior professionals (for career growth).
- Assisted in organizing engineering society events within the local community.